

Class 2

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Today we'll explore the gapminder data. It contains life expectancy, population, and GDP per capita for countries every five years from 1952 to 2007. The gapminder data comes from the Gapminder Foundation: <https://www.gapminder.org/data/> made famous by Hans Rosling's TED talks. We'll continue our exploration using R to understand and visualize data.

Part 1: Getting to Know Gapminder

Make sure you have opened the Rproject first.

Today's data comes packaged inside R itself. Instead of loading a file, we install a package once, then load it with `library()` each session.

Packages are like bonus material for R. We will use them all the time. Installing puts it on your computer. Library opens it (like a book) so everytime you use it you need library in your code first.

```
#install.packages("gapminder") #only run this once ever - then put the # back in front
library(gapminder)
data("gapminder") #Puts into our environment

#install.packages("dplyr") #We'll use this package later, but always a good idea to load pack
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

Looking over the data using commmands from yesterday

(hint: use code from yesterday)

```
head(gapminder)
```

```
# A tibble: 6 x 6
  country    continent  year lifeExp      pop gdpPercap
  <fct>      <fct>    <int>  <dbl>   <int>   <dbl>
1 Afghanistan Asia      1952   28.8  8425333    779.
2 Afghanistan Asia      1957   30.3  9240934    821.
3 Afghanistan Asia      1962   32.0 10267083    853.
4 Afghanistan Asia      1967   34.0 11537966    836.
5 Afghanistan Asia      1972   36.1 13079460    740.
6 Afghanistan Asia      1977   38.4 14880372    786.
```

```
tail(gapminder)
```

```
# A tibble: 6 x 6
  country    continent  year lifeExp      pop gdpPercap
  <fct>      <fct>    <int>  <dbl>   <int>   <dbl>
1 Zimbabwe Africa      1982   60.4  7636524    789.
2 Zimbabwe Africa      1987   62.4  9216418    706.
3 Zimbabwe Africa      1992   60.4 10704340    693.
4 Zimbabwe Africa      1997   46.8 11404948    792.
5 Zimbabwe Africa      2002   40.0 11926563    672.
6 Zimbabwe Africa      2007   43.5 12311143    470.
```

```
summary(gapminder)
```

	country	continent	year	lifeExp
Afghanistan:	12	Africa	:624	Min. :1952
Albania	: 12	Americas:	300	Min. :23.60
Algeria	: 12	Asia	:396	1st Qu.:1966
Angola	: 12	Europe	:360	1st Qu.:48.20
Argentina	: 12	Oceania	: 24	Median :1980
				Median :60.71
				Mean :59.47
				Mean :70.85

```

Australia : 12                      Max. :2007   Max. :82.60
(Other)    :1632
      pop      gdpPercap
Min. :6.001e+04 Min. : 241.2
1st Qu.:2.794e+06 1st Qu.: 1202.1
Median :7.024e+06 Median : 3531.8
Mean :2.960e+07 Mean : 7215.3
3rd Qu.:1.959e+07 3rd Qu.: 9325.5
Max. :1.319e+09 Max. :113523.1

```

```
dim(gapminder) #Count of how many observations and variables
```

```
[1] 1704    6
```

```
str(gapminder) #This tells us what kind of data R thinks each variable is - we'll talk about
```

```

tibble [1,704 x 6] (S3: tbl_df/tbl/data.frame)
 $ country : Factor w/ 142 levels "Afghanistan",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ continent: Factor w/ 5 levels "Africa","Americas",...: 3 3 3 3 3 3 3 3 3 3 ...
 $ year      : int [1:1704] 1952 1957 1962 1967 1972 1977 1982 1987 1992 1997 ...
 $ lifeExp   : num [1:1704] 28.8 30.3 32 34 36.1 ...
 $ pop       : int [1:1704] 8425333 9240934 10267083 11537966 13079460 14880372 12881816 1386...
 $ gdpPercap: num [1:1704] 779 821 853 836 740 ...

```

```
length(unique(gapminder$country)) #gives a count of unique countrys
```

```
[1] 142
```

Checking in (with a partner)

- 1) How many countries are in the dataset?
- 2) What is the average life expectancy?
- 3) What is the median GDP per capita? Average? Bonus:
 - a) Create a scatterplot with the relationship between population and Life Expectancy. Add a trend line b)What country has the highest life expectancy?

It is best practice to “render as you go.” Try rendering now to make sure you don’t have errors in your code.

Country and continent are not continuous(numeric in R speak) variables. They are categorical. We can get counts in each category.

frequency tables:

```
table(gapminder$continent) #gives count of different categories
```

Africa	Americas	Asia	Europe	Oceania
624	300	396	360	24

```
table(gapminder$year)
```

1952	1957	1962	1967	1972	1977	1982	1987	1992	1997	2002	2007
142	142	142	142	142	142	142	142	142	142	142	142

```
prop.table(table(gapminder$continent)) #gives percentages
```

Africa	Americas	Asia	Europe	Oceania
0.36619718	0.17605634	0.23239437	0.21126761	0.01408451

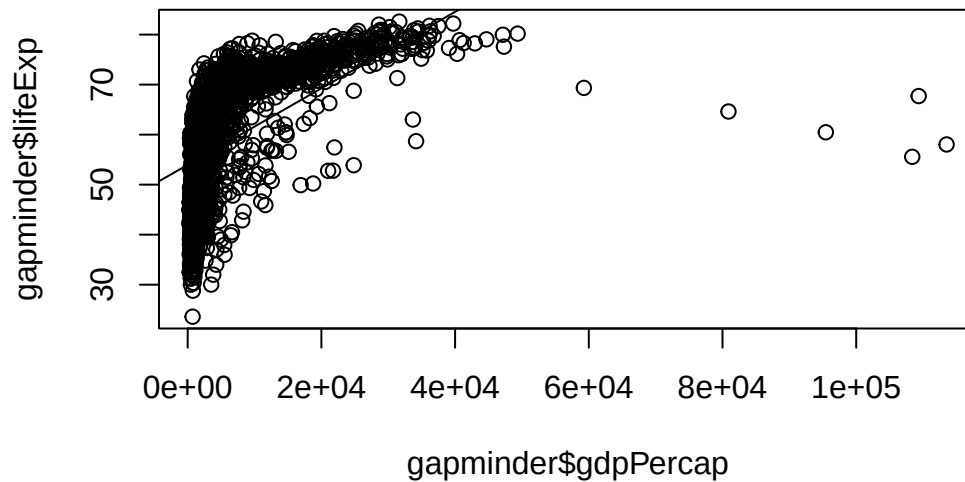
I use this command all the time to get a sense of the data.

Check in: What percentage of the observations in the dataset are from Asia?

“

Plot the relationship between GDP per capita and life expectancy

```
plot(gapminder$gdpPercap, gapminder$lifeExp) #scatterplot (X then Y)
abline(lm(gapminder$lifeExp~gapminder$gdpPercap)) #add trend line (Y then X)
```



Notice how we always put the explanatory variable on the X axis(horizontal) and the outcome variable on the Y axis (vertical)

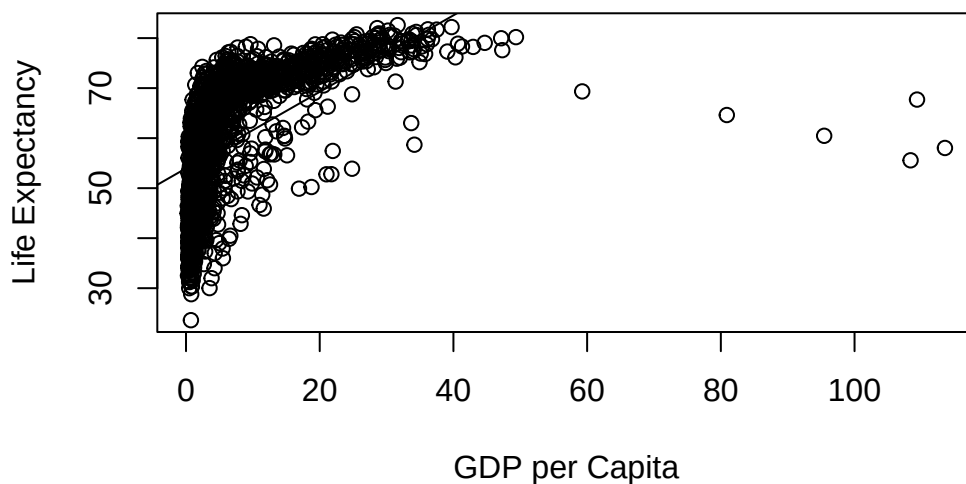
GDP per capita hard to read - change unit to thousands

```
gapminder$gdpPercap <- gapminder$gdpPercap/1000
```

Make plot look a bit more professional

```
plot(gapminder$gdpPercap, gapminder$lifeExp, main="Relationship between Wealth and Life Expectancy",
      abline(lm(gapminder$lifeExp~gapminder$gdpPercap))) #and boom, you've already ran your 2nd regression
```

Relationship between Wealth and Life Expectancy



Does there appear to be a relationship in the sample?

What about with just Asia? There are multiple ways to do this:

We can create a new dataset ie a subset(I almost always choose this option)

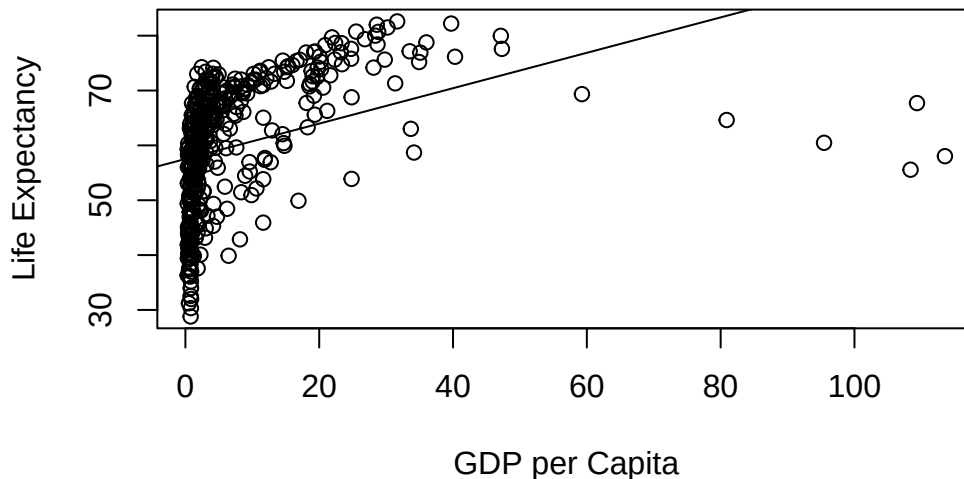
```
table(gapminder$continent)
```

Africa	Americas	Asia	Europe	Oceania
624	300	396	360	24

```
asia <- gapminder[which(gapminder$continent=="Asia"),]  
asia <- filter(gapminder, continent=="Asia") # This does the same thing using the dplyr pack
```

```
plot(asia$gdpPercap, asia$lifeExp, main="Relationship between Wealth and Life Expectancy (Asia)",  
     abline(lm(asia$lifeExp~asia$gdpPercap))) #and boom, another regression
```

Relationship between Wealth and Life Expectancy (Asia)



Exclude Europe

```
noteuro <- gapminder[which(gapminder$continent!="Europe"),] # the != does not equal  
noteuro <- filter(gapminder, continent!="Europe")
```

Exclude the super rich - use greater than or less than

```
notrich <- gapminder[which(gapminder$gdpPercap < 50),] # less than 50,000 - this drops oil-rich countries
notrich <- filter(gapminder, gdpPercap < 50)
```

Check in: Try creating a scatterplot with just the 1) Africa OR 2) country-years where life expectancy is less than 70.

#Looking at the data over time: Time to look at something a bit more fun. How has life expectancy changed across the world since the 1950s?

Mean life expectancy?

```
mean(gapminder$lifeExp)
```

```
[1] 59.47444
```

```
min(gapminder$lifeExp)
```

```
[1] 23.599
```

```
max(gapminder$lifeExp)
```

```
[1] 82.603
```

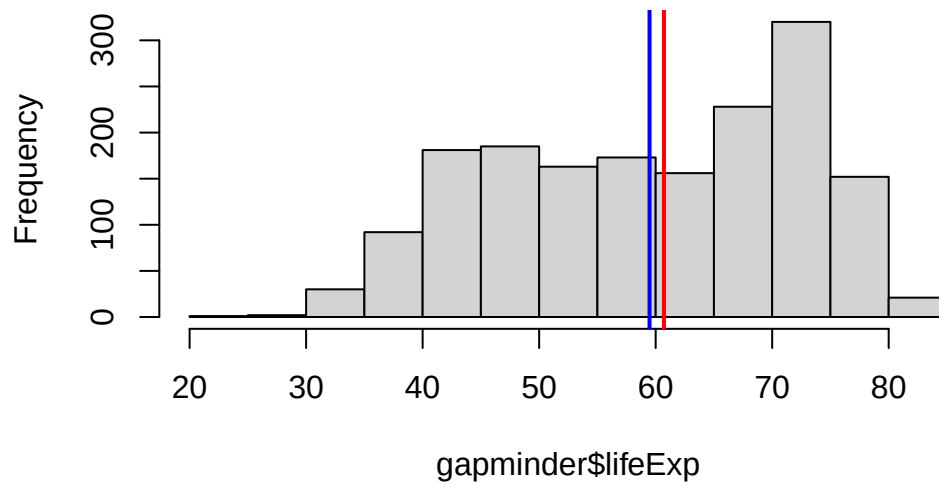
```
range(gapminder$lifeExp)
```

```
[1] 23.599 82.603
```

Create a histogram for life expectancy

```
hist(gapminder$lifeExp)
abline(v=mean(gapminder$lifeExp), col="blue", lwd=2)#adds a vertical(v) line for the mean
abline(v=median(gapminder$lifeExp), col="red", lwd=2)#a a line for mediant
```

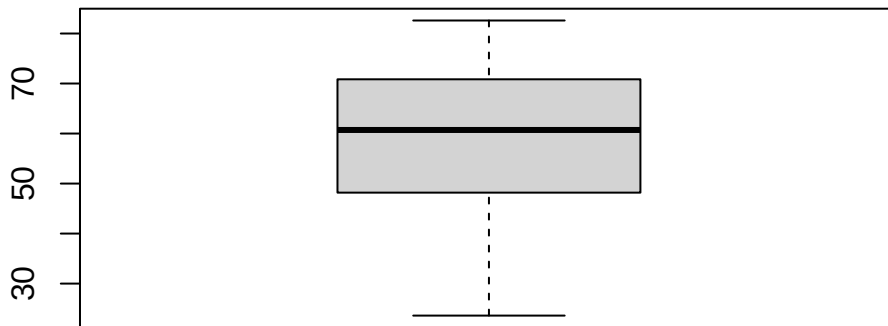
Histogram of gapminder\$lifeExp



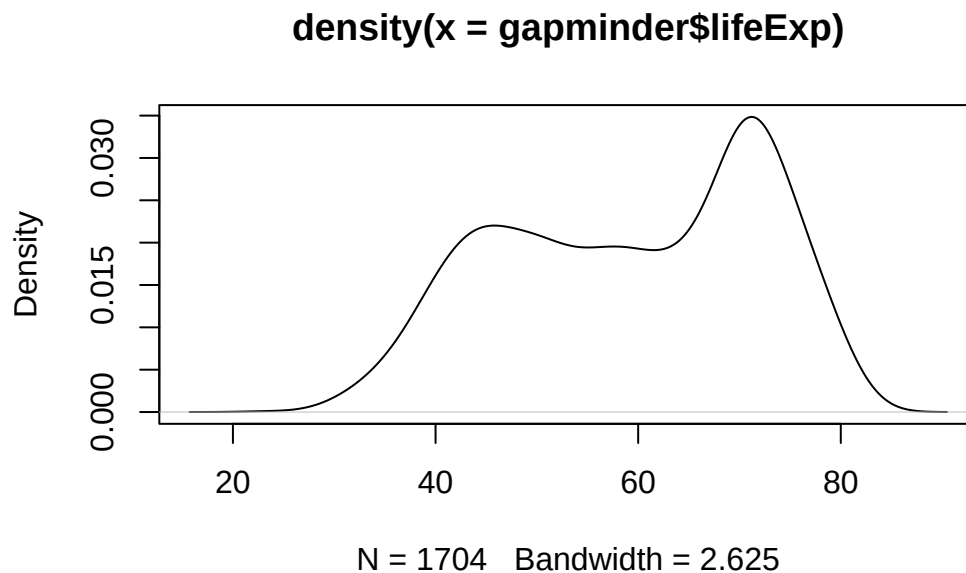
abline gives us a line. v is for vertical

Density plots and boxplots show us similar information

```
boxplot(gapminder$lifeExp)
```



```
plot(density(gapminder$lifeExp))
```



```
#How can we find out which country-years are at the very bottom?
```

Check: How many observations are from Africa and how many are from Europe

```
mean(gapminder$lifeExp[gapminder$continent=="Africa"]) #average for Africa
```

```
[1] 48.86533
```

```
mean(gapminder$lifeExp[gapminder$continent=="Europe"]) #average for Europe
```

```
[1] 71.90369
```

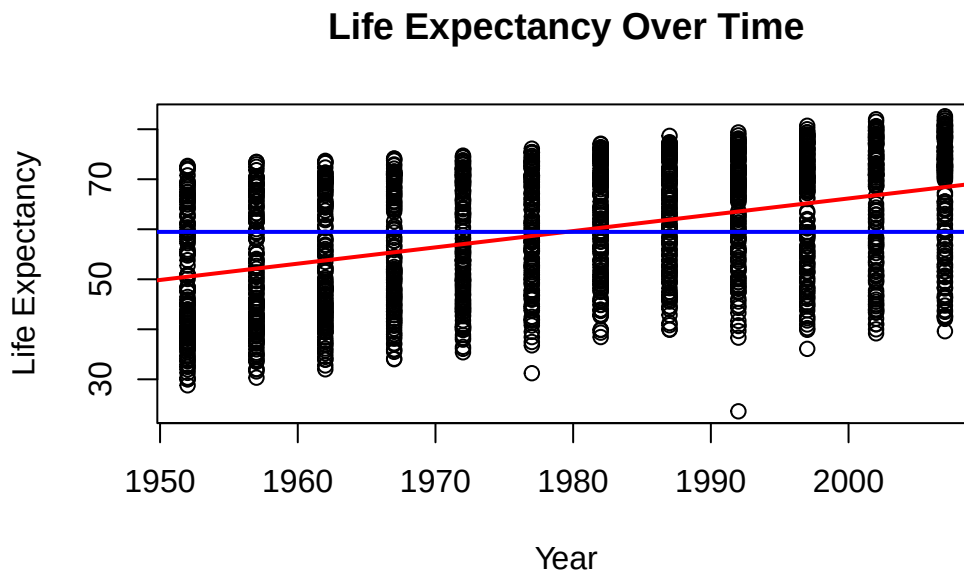
```
mean(asia$lifeExp)
```

```
[1] 60.0649
```

```
#find the mean for the Americas
```

Let's make a simple scatterplot with year on the horizontal axis, and life expectancy on the vertical axis. A regression allows us to look at the relationship between variables linear regression model, and you're looking at the effect of the IV on the DV

```
plot(gapminder$year, gapminder$lifeExp, ylab = "Life Expectancy", xlab = "Year", main = "Life Expectancy Over Time")
abline(lm(gapminder$lifeExp ~ gapminder$year), col="red", lwd=2) # What type of relationship
abline(h=mean(gapminder$lifeExp), col="blue", lwd=2)
```



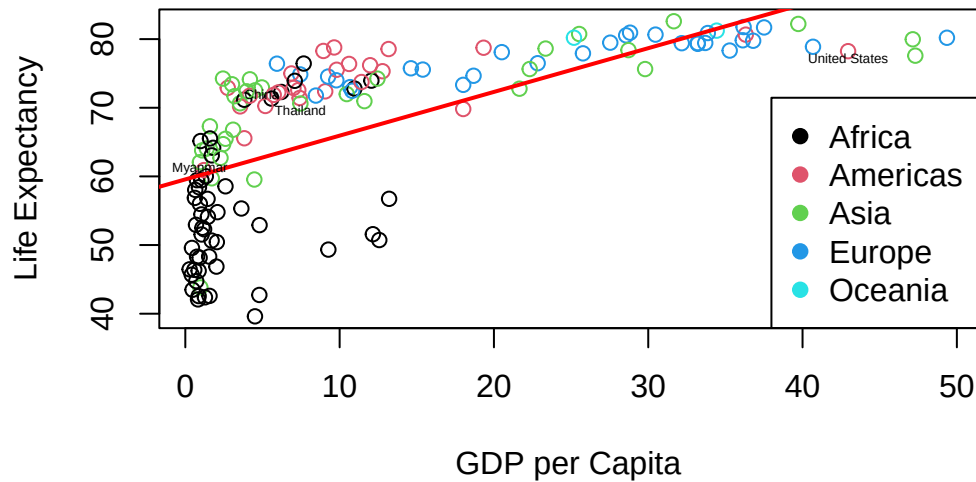
That is a bit overwhelming lets just look at 2007 - ie lets create a subset

```
g2007 <- gapminder[which(gapminder$year==2007),] #just using base R

#New Graph
plot(g2007$gdpPercap, g2007$lifeExp, ylab = "Life Expectancy", xlab = "GDP per Capita", main = "Life Expectancy vs GDP per Capita")
abline(lm(g2007$lifeExp ~ g2007$gdpPercap), col="red", lwd=2)
legend("bottomright", legend = levels(factor(g2007$continent)), pch=19, col = factor(levels(g2007$continent)))

#I want to add labels for specific countries
g2007$stars <- ifelse(g2007$country=="United States" | g2007$country=="China" | g2007$country=="Russia", 1, 0)
text(g2007$gdpPercap, g2007$lifeExp-1, g2007$stars, cex=.4) #add text to location
```

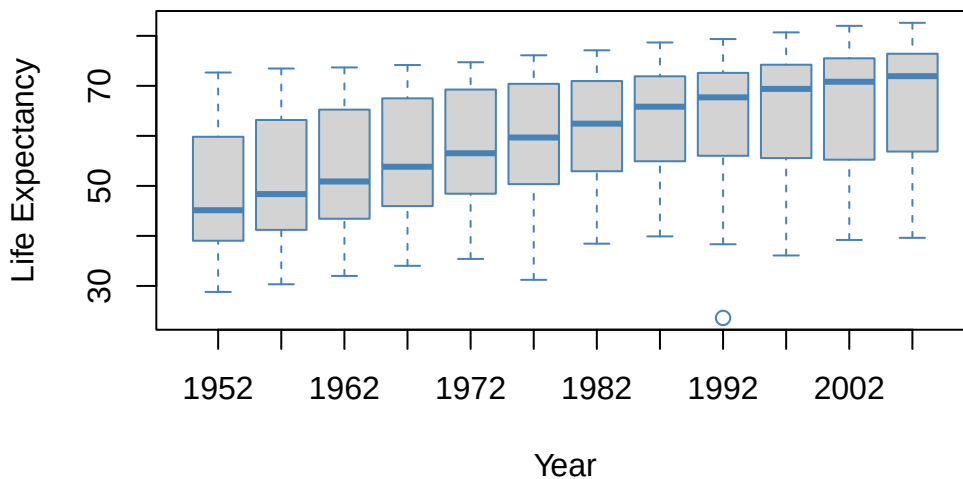
Wealth and Life Expectancy in 2007



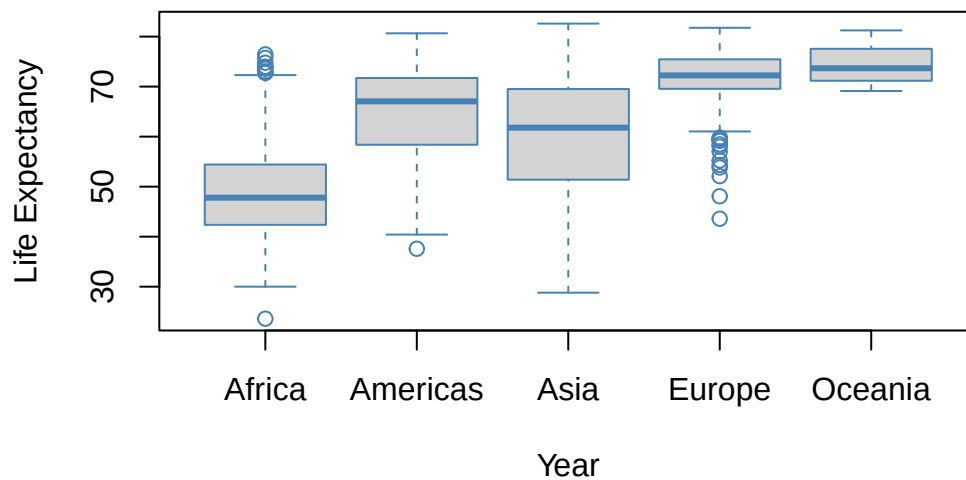
Check: Can you create the same for 1952

A boxplot for each year can show us how the distribution has changed over time

```
boxplot(gapminder$lifeExp~gapminder$year, xlab = "Year", ylab="Life Expectancy", border = "s
```



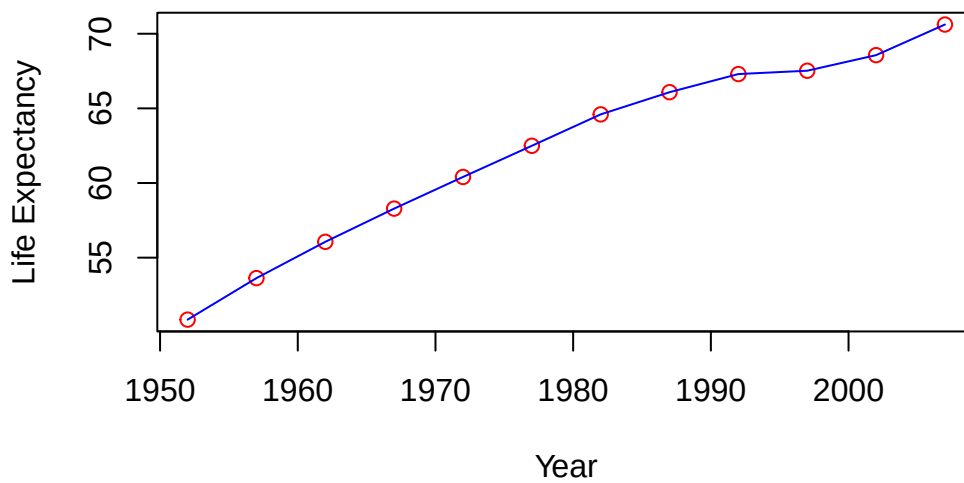
```
boxplot(gapminder$lifeExp~gapminder$continent, xlab = "Year", ylab="Life Expectancy", border
```



Lets look at changes in thailand over time

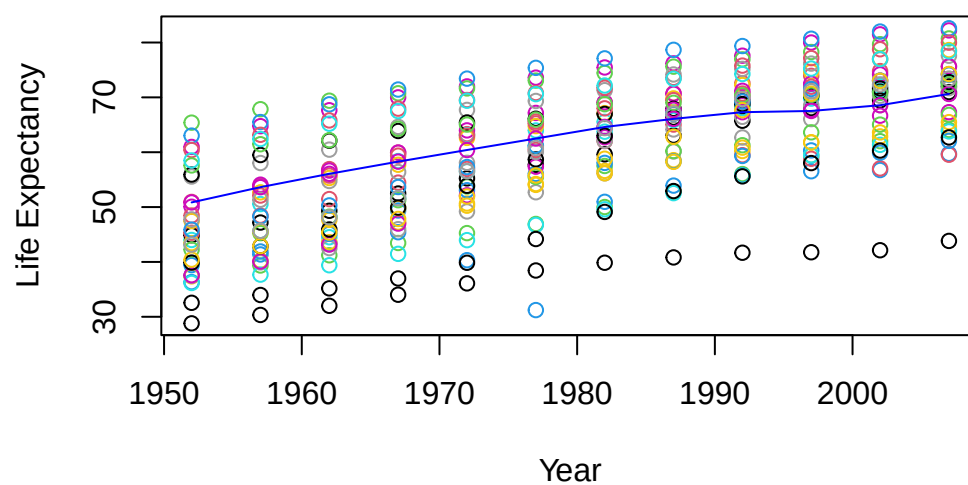
```
thai <- filter(gapminder, country=="Thailand")
plot(thai$year, thai$lifeExp, xlab="Year", ylab="Life Expectancy", col="red", main="Thailand")
lines(thai$year, thai$lifeExp, col="blue")
```

Thailand Life Expectancy



```
plot(asia$year, asia$lifeExp, xlab="Year", ylab="Life Expectancy", col=gapminder$country, main="Thailand")
lines(thai$year, thai$lifeExp, col="blue")
```

Thailand Life Expectancy



Check: Do the same above for a different country

I've clearly had too much fun. Your turn to make something interesting.

Since we've created some new variables we may want to save our data:

```
write.csv(g2007, "g2007.csv") #writes a csv to your working directory that you can open in e
save(g2007, file="g2007.Rdata") #saves Rdata format to your working directory
```